
How ChronoHive Works

System architecture and evaluation guide

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A plain-language tour of the ChronoHive evaluation system: what each part does, how a request flows through it, and how to read your evaluation numbers honestly.

The big picture

ChronoHive is a **storage-admission kernel**: it decides which I/O work gets admitted, when, and at what quality-of-service level — so that coordinated workloads stall less than uncoordinated ones. The evaluation package lets you drive the **real kernel** over HTTPS and measure that coordination behavior. The storage hardware underneath is **simulated**; the decisions are not.

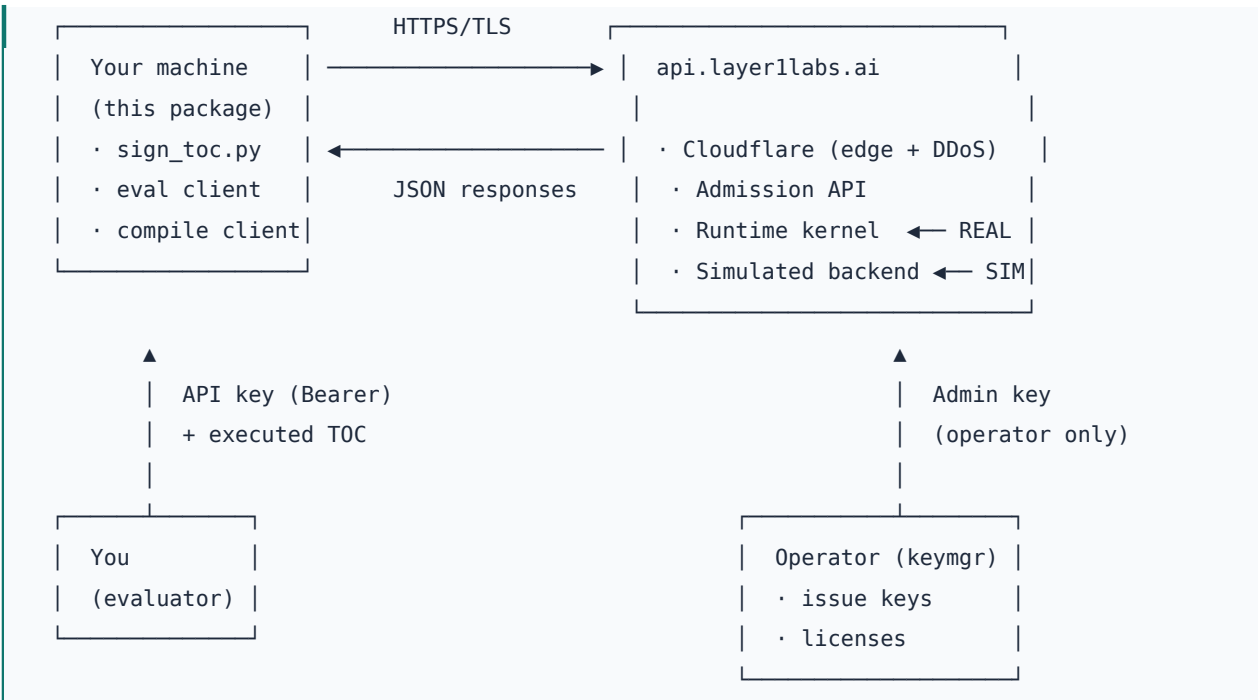


Figure 1 — System overview. You talk to one HTTPS API. Behind it, the real Runtime kernel makes every grant/refuse decision; a simulated storage backend, contention model, and storage API surface stand in for hardware. The operator side (key issuance, licenses) is separate and never exposed to evaluators.

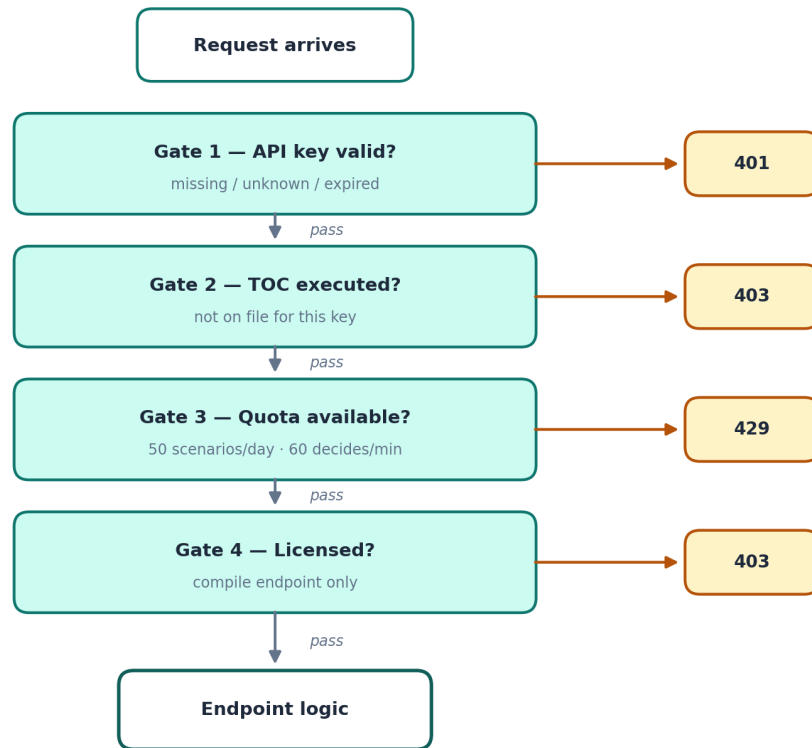
There are exactly three kinds of component in the system, and the honesty of your evaluation depends on not confusing them:

Component	Real or simulated?	What it means for you
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Runtime kernel (admission decisions)	Real	Every grant/refuse comes from the kernel's capacity discipline. This is what you are evaluating.
Storage backend, contention model, storage API surface	Simulated	Stand-ins for hardware. They let the kernel's decisions play out observably without vendor hardware.
Your numbers (stall reduction, makespan)	Simulation outcomes	Valid for comparing coordination behavior (kernel vs uncoordinated). Not hardware measurements. Do not present them as such.

How a request is gated

Every request passes the same four gates before it reaches any endpoint logic. The first failure wins, in this order:

How a request is gated — first failure wins, in order

Request gating

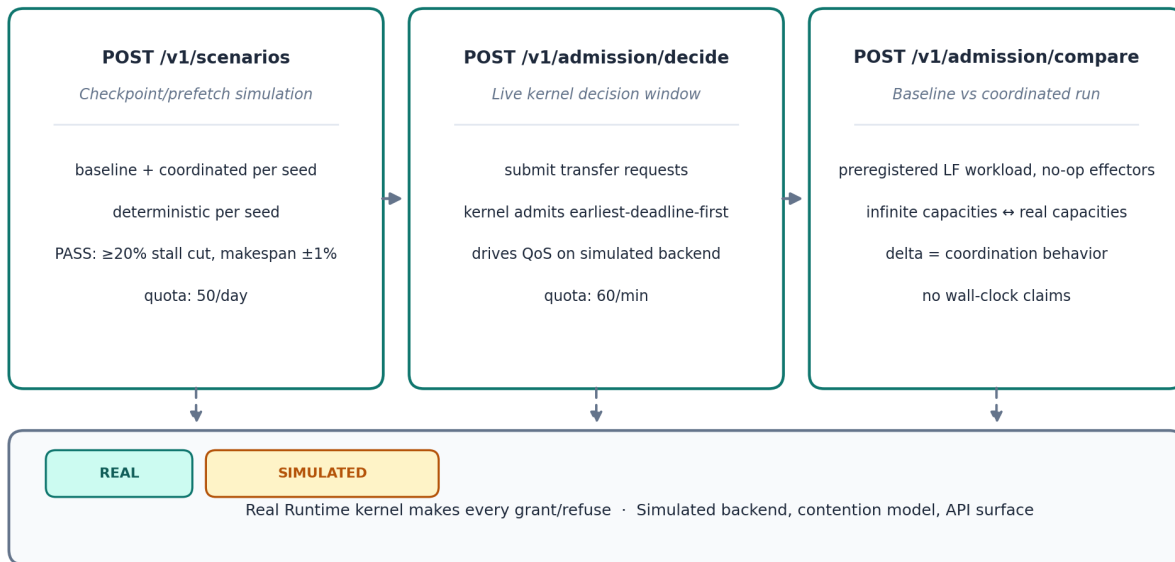
Figure 2 — Request gating. **401** = your key is missing, unknown, or expired. **403** **toc_acceptance_required** = execute the TOC first (see Quickstart). **429** = quota exhausted (50 scenarios/day, 60 decides/minute). **POST /v1/lf/compile** has one extra gate: your key needs a **compile license**, issued by the operator alongside your key.

How evaluation works

Evaluation answers one question: **does kernel-coordinated admission stall less than uncoordinated admission?** Four endpoints, one story:

How evaluation works — one question, three endpoints

Does kernel-coordinated admission stall less than uncoordinated admission?



GET /v1/scenarios/{id} fetches any stored result for your records

Evaluation flow

Figure 3 — The evaluation flow.

- 1. `POST /v1/scenarios` — the checkpoint/prefetch simulation.** Runs the same workload twice per seed: once uncoordinated (baseline — everything the schedule asks for is admitted) and once kernel-coordinated (the real kernel's admission discipline applies). Deterministic per seed: the same seed always produces the same result. A scenario **passes** when the coordinated run cuts stalls by $\geq 20\%$ with makespan within 1% of baseline. Quota: 50 runs per key per rolling day.
- 2. `POST /v1/admission/decide` — the live kernel decision path.** You submit transfer requests for one admission window; the real kernel admits them earliest-deadline-first against the provisioned write/read pipes and drives QoS levels on the simulated backend from its grant decisions. This is the kernel thinking out loud, one window at a time. Quota: 60 calls per key per minute.
- 3. `POST /v1/admission/compare` — baseline vs coordinated, head to head.** A preregistered Lingua Franca workload runs twice through the real Runtime kernel with no-op effectors: once with effectively infinite capacities (uncoordinated baseline) and once with the specification's real capacities (coordinated). The delta is the kernel's coordination

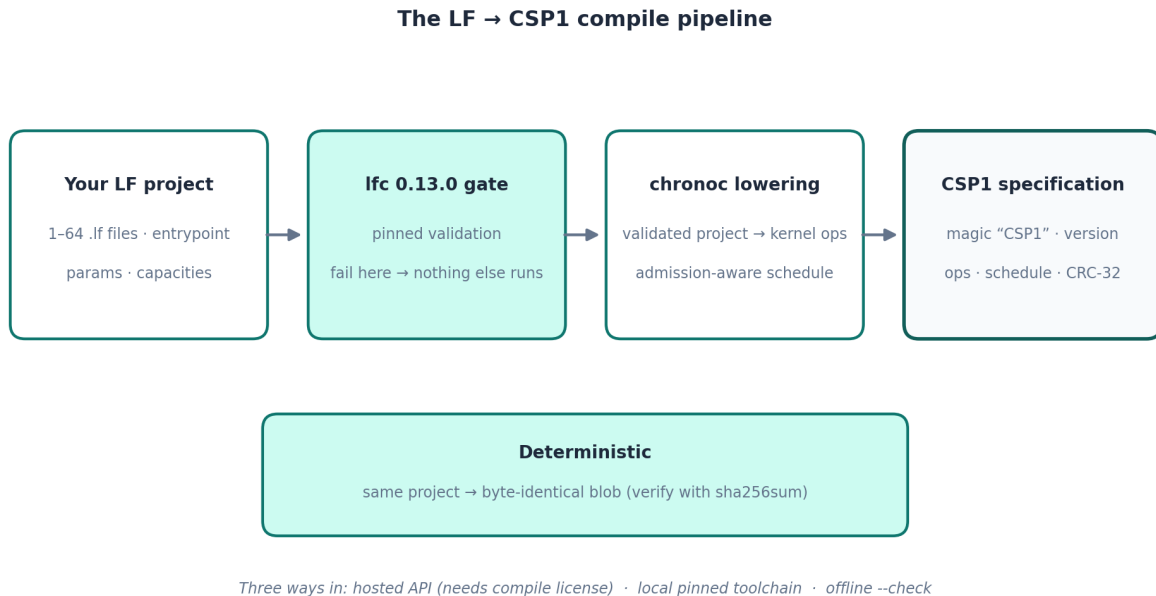
behavior — who is admitted when, what retries, what drops. Admission decisions are real; no I/O timing is simulated or measured, so no wall-clock speedup is claimed.

4. ``GET /v1/scenarios/{id}`` fetches any stored scenario result back for your records.

The worked example (`examples/eval_walkthrough.py`) runs this whole flow — TOC → scenario → live decisions → result fetch — in one command. See the Quickstart in `../README.md`.

How compilation works

Lingua Franca workloads compile to `.cspec` v1 specifications — the artifact the engine executes. Compilation is a pipeline with a hard validation gate up front:



Compile pipeline

Figure 4 — The LF → CSP1 compile pipeline.

1. **Your LF project** — a complete file map (1-64 `.lf` files), an endpoint, integer params, and capacity declarations.
2. **``lfc` 0.13.0 validation gate`** — the pinned Lingua Franca compiler checks the project. If it fails here, nothing else runs.
3. **``chronoc` lowering`** — the ChronoHive compiler lowers the validated project to the

kernel's operation set (e.g. `c.admit_checkpoint`) and serializes the **CSP1 specification**: magic "CSP1", version, step table, string table, capacities, operations, schedule, LF→op bindings, source provenance (SHA-256, toolchain versions), and a CRC-32 trailer.

4. Deterministic — the same project always produces the byte-identical specification. Verify with `sha256sum`.

You can compile three ways: the hosted API (`clients/compile_client.py` with [api-url](#), needs a compile license), the local pinned toolchain (`local`), or the offline self-test (`check, no network`). Compiling is optional for evaluation — the admission API needs neither `lfc` nor `chronoc`. See [LF_TOOLCHAIN.md](#) for the toolchain pins and [../README.md](#) for the commands.

Reading your numbers honestly

- **Do compare:** coordinated vs uncoordinated within a scenario or compare run. That delta is the kernel's doing, and it is real.
- **Do not compare:** evaluation numbers against hardware measurements, or present them as hardware performance. The backend is simulated; every response says so (`"simulated_backend": true`).
- **Determinism is your friend:** scenarios are deterministic per seed, and compiles are byte-deterministic. If a number changes, something in the inputs changed — find it before concluding anything.

Endpoint map

Endpoint	What it does	Gate notes
GET /v1/health	Liveness + version + TOC hash	Public
GET /v1/toc	Canonical Terms of Confidentiality	Public
POST /v1/toc/accept	Submit signed TOC acceptance	Key required
POST /v1/scenarios	Run admission simulation (baseline + coordinated)	Key + TOC; 50/day
GET /v1/scenarios/{id}	Fetch a stored scenario result	Key + TOC

POST
/v1/admission/decide

Live kernel decisions for one window

Key + TOC; 60/min

POST
/v1/admission/compare

Baseline vs coordinated workload run

Key + TOC

POST /v1/lf/compile

Compile an LF project to a CSP1
specification

Key + TOC + compile
license

Full request/response contracts are in **API.md**.